SOUVIK KONAR

Chennai, Tamil Nadu, India | +91 8927172288 | souvikkonar2005@gmail.com | linkedin.com/in/souvikkonar99 | github.com/SOUVIKKONAR

SUMMARY

Final-year Computer Science and Engineering (Information Technology) undergraduate at SRM Institute of Science and Technology, graduating May 2027, with hands-on experience training, evaluating, and deploying machine learning and deep learning models. Built a custom-trained YOLOv8 computer vision model deployed on edge hardware for real-time dual-class object detection, and co-authored a published IEEE research paper on an IoT automation system. Proficient in Python, PyTorch, TensorFlow, and OpenCV, with practical exposure to dataset preparation, model tuning and evaluation, and deploying trained models into working applications via REST APIs and real-time inference pipelines. Complements this with full-stack development experience (Django REST Framework, React) that enables end-to-end ownership of ML-powered products, from data pipeline to production deployment.

TECHNICAL SKILLS

Machine Learning & Deep Learning: Model Training, Tuning & Evaluation, Computer Vision (YOLOv8), Data Preprocessing, Feature Engineering
ML/DL Frameworks & Libraries: PyTorch, TensorFlow, OpenCV, NumPy, Pandas
Languages: Python, Java, JavaScript, C, C++, SQL
Model Deployment & APIs: REST APIs, Flask-SocketIO, Django REST Framework, OpenAI API, Prompt Engineering
Cloud & Databases: AWS Lambda, Render, MySQL, MongoDB, SQLite
Data Collection & IoT: ESP32, Arduino IDE, MQTT, Sensor Integration (real-time data streams)
Tools & Practices: Git, GitHub, Object-Oriented Design, Technical Documentation, Problem Solving

INTERNSHIPS

Revolite Infotech Pvt. Ltd. — Python Full Stack Intern *Jun 2026 – Jul 2026*
Chennai, India | 45-day internship | Certificate

- Built Cadence, a full-stack employee management platform, as the primary project of the internship, using Django REST Framework and React
- Designed and developed RESTful APIs, relational database models, and JWT-based authentication under mentorship in a Python full-stack environment
- Recognized for sincere, dedicated, and result-oriented performance throughout the internship

Steel Authority of India Limited (SAIL) — Vocational Trainee *Dec 2025*
Durgapur Steel Plant, West Bengal | 11-day training | Certificate

- Completed vocational training across Blast Furnace, Coke Ovens, RMHP, Sinter Plant, and SMS departments
- Gained hands-on exposure to industrial manufacturing operations and plant-level engineering workflows

PROJECTS

RoadSense — Real-Time Pothole & Waste Detection System

- Prepared and labeled a custom image dataset, then trained and evaluated a YOLOv8 computer vision model for dual-class (pothole and waste) object detection deployed on ESP32-CAM/Arduino edge hardware
- Tuned model confidence thresholds and built server-side input validation to filter low-confidence predictions and reduce false positives
- Built a Flask-SocketIO backend to serve real-time model inference results to a React and Leaflet geospatial dashboard

IntelliCode — AI-Assisted Code Editor

- Built an AI-assisted code editor using React, Monaco Editor, and an Express.js backend integrated with the OpenAI API
- Designed REST endpoints and prompt flows for automated code debugging, explanation, and optimization, returning structured JSON responses
- Implemented request validation and configurable API routing for reliable, production-ready client-server communication

Cadence — Employee Management Platform

- Built a full-stack employee management platform using Django REST Framework and React with JWT-based authentication and role-based access control
- Implemented attendance, leave, and payroll tracking modules backed by RESTful APIs and a relational database
- Deployed the application on Render with environment-based configuration for production readiness

RESEARCH PUBLICATION

Intelligent Home Automation System With Weather Forecasting and Alert Integration *Feb 2025 – Dec 2025*
13th IEEE International Conference | IEEE Xplore

- Developed NexHome, an IoT home automation system using ESP32 and multiple environmental sensors, applying rule-based analysis to sensor data streams for automated decision logic
- Implemented automation logic with Spring Boot and Telegram Bot API for real-time alerts
- Achieved a 30% improvement in energy efficiency through optimized IoT communication protocols

EDUCATION

B.Tech in Computer Science and Engineering with Information Technology *Expected May 2027*
SRM Institute of Science and Technology, Kattankulathur, Chennai | GPA: 8.83/10.00

CERTIFICATES & ACHIEVEMENTS

All certificates and achievements: Google Drive folder

- Intel Unnati AIoT Program (May 2025) — Trained in AIoT systems integrating AI and IoT
- SAP Generative AI Developer (Mar 2026 – Mar 2027) — Developed Generative AI applications using SAP BTP
- NPTEL Certifications: Programming in Java, Design and Analysis of Algorithms, Database Management Systems (2024–2025)
- Hackathons: Bitathon 2024 (Qualified 3/4 rounds), Smart India Hackathon 2024 (Internal Edition)